

TITLE

IoT-Based Smart Greenhouse Monitoring System Using ESP32

ABSTRACT

This project presents an IoT-based Smart Greenhouse Monitoring System that uses an ESP32 microcontroller and various environmental sensors to track key conditions such as temperature, humidity, soil moisture, and light intensity. The data is transmitted in real-time via Wi-Fi to a mobile application, enabling farmers or gardeners to remotely monitor the health of the greenhouse. The system is designed to be cost-effective, scalable, and energy-efficient, providing an innovative solution to improve productivity in agriculture. Unlike previous systems, our design integrates wireless monitoring with mobile technology for continuous and remote data visibility.

INTRODUCTION

❖ Problems / Questions Our Project Will Address :

- Traditional greenhouse farming relies heavily on manual monitoring, which is time-consuming and prone to error.
- Environmental parameters like temperature, humidity, soil moisture, and light intensity must be constantly monitored for optimal crop growth.
- Lack of real-time data and remote monitoring leads to crop stress and reduced yield.

❖ What Our Project Does :

- Our project offers an automated, real-time monitoring system for greenhouses using IoT technologies.
- It continuously tracks temperature, humidity, soil moisture, and light intensity using low-cost sensors.
- It transmits the live data wirelessly to a mobile application using ESP32's built-in Wi-Fi, allowing users to monitor from anywhere.

❖ Importance to the Field :

- The system supports precision agriculture, helping farmers make data-driven decisions.
- It minimizes labor, improves resource usage (especially water), and maximizes crop yield.
- Promotes the adoption of smart farming in small and medium-scale agricultural setups.

LITERATURE SURVEY

<u>Author</u>	<u>Title</u>	<u>Type / Year</u>	<u>Summary / Key contributions</u>
Cangqing Wang	Intelligent Agricultural Greenhouse Control System Based on Internet of Things and Machine Learning	Preprint / Feb 2024	Proposes a greenhouse control architecture integrating IoT with machine learning to improve crop growth, resource efficiency, and environmental control.
Pinar Kirci, Erdinc Ozturk, Yavuz Celik	A Novel Approach for Monitoring of Smart Greenhouse and Flowerpot Parameters and Detection of Plant Growth with Sensors	Journal Article / Oct 2022	Describes a smart greenhouse prototype managing different crops (e.g., vegetables, flowers) and demonstrates ~16.5% water savings and healthier plant growth compared to outdoor pots.
B. Rama Devi et al. (conference)	A Comprehensive Review of the Application of Greenhouse Using the Internet of Things	Conference Paper / Mar 2024	Reviews IoT applications in greenhouse farming—covering sensors, communication protocols, challenges, and future opportunities.
M.S. Farooq et al. (Web of Things review)	Web of Things and Trends in Agriculture: A Systematic Literature Review	Preprint / Jun 2023	Examines the adoption of Web of Things (WoT) for agriculture—highlighting IoT integration, interoperability, and emerging future directions.
Arcel Christian H. Austria et al.	Development of IoT Smart Greenhouse System for Hydroponic Gardens	Preprint / May 2023	Presents a mobile-linked IoT system for hydroponic greenhouses, monitoring pH, light, temperature, humidity; evaluated via ISO 9126 quality metrics, scoring “Very Good.”
Multiple authors (IEEE Trans. Veh. Technol. 2022)	IoT Based Smart Greenhouse Framework and Control Strategies for Sustainable Agriculture	Journal / 2022	Proposes an IoT-driven framework with control algorithms aimed at promoting sustainable greenhouse operations.
Multiple authors (Computers and Electronics in Agriculture, 2022)	Applications of IoT for optimized greenhouse environment and resources management	Journal / 2022	Explores IoT applications for optimizing greenhouse microclimate and resource use.
Multiple authors (IEEE Trans. Veh. Technol. 2023)	Battery-Free LiFi-Enabled IoT for Sustainable Greenhouses	Journal / 2023	Introduces a LiFi-based battery-free IoT system as a sustainable alternative for greenhouse connectivity.

➤ **1. Project** : “Greenhouse Monitoring System Using Arduino and GSM”

Overview :

This system used Arduino UNO with GSM module to send SMS alerts when temperature or humidity crossed threshold values.

Research Gap :

The project lacked real-time continuous monitoring and remote visualization. SMS-based systems provide limited and delayed updates.

Our Improvement :

By using ESP32 with Wi-Fi, our project enables continuous, real-time monitoring and instant access to live data via a mobile application.

➤ **2. Project** : “Automated Greenhouse Monitoring System Using Raspberry Pi”

Overview :

Raspberry Pi was used with sensors to collect data and control actuators. Data was displayed using an LCD and local GUI.

Research Gap :

High cost and power consumption; also required more complex setup and operating system, which may not be ideal for rural or small-scale users.

Our Improvement :

We use ESP32, which is low-power, low-cost, and easy to program using Arduino IDE, making the system more affordable and accessible for small farmers or students.

➤ **3. Project** : “Smart Greenhouse Using Arduino and Bluetooth Communication”

Overview :

This project used Arduino and Bluetooth module to send greenhouse parameter data to an Android app within short-range distance.

Research Gap :

Limited range of Bluetooth restricted its use to within 10–15 meters, making it unsuitable for remote monitoring.

Our Improvement :

Our project uses ESP32's Wi-Fi capability to send data to the cloud or mobile app, allowing remote access from any location with internet, thus eliminating the distance limitation.

➤ **4. Project :** “IoT-Based Greenhouse System Using NodeMCU and Thingspeak”

Overview :

NodeMCU (ESP8266) was used with sensors to upload data to Thingspeak for cloud visualization.

Research Gap :

The project only focused on web-based dashboard and lacked mobile app integration, making it less intuitive and harder to interact with for non-technical users.

Our Improvement :

Our system integrates with a mobile app (e.g., Blynk) for user-friendly, real-time mobile dashboard, ensuring easy interaction for farmers, students, or hobbyists.

OBJECTIVES OF SMART GREENHOUSE PROJECT :

1. To monitor key environmental parameters such as:
 - Temperature
 - Humidity
 - Soil moisture
 - Light intensity
2. To automate greenhouse control systems based on sensor data by:
 - Turning on/off a fan to regulate temperature and humidity
 - Turning on/off a water sprayer based on soil moisture.
 - Controlling LED grow lights based on ambient light conditions

3. To implement real-time data transmission from the greenhouse to a mobile application using Wi-Fi (ESP32 or ESP8266).
4. To design an energy-efficient and cost-effective embedded system for sustainable greenhouse management.
5. To enhance remote monitoring and control capabilities by integrating IoT features.
6. To provide a user-friendly mobile interface for:
 - Live data viewing
 - System alerts or notifications
 - Manual override of automation (if needed)

HYPOTHESIS

"If we monitor critical greenhouse parameters in real-time using IoT technologies, then we can significantly improve crop health, reduce manual labor, and enhance agricultural productivity in a cost-effective manner."

▪ **Primary Hypothesis :**

If we monitor critical greenhouse parameters in real-time using IoT technologies, then we can significantly improve crop health, reduce manual labor, and enhance agricultural productivity in a cost-effective manner.

▪ **Sensor Integration Hypothesis :**

If temperature, humidity, soil moisture, and light intensity are continuously measured using sensors (DHT11, soil moisture sensor, and LDR), then the greenhouse environment can be more precisely regulated for optimal plant growth.

▪ **Connectivity Hypothesis :**

If ESP32 is used to transmit live environmental data over Wi-Fi to a mobile app, then farmers and users can monitor the greenhouse conditions remotely without being physically present.

▪ **Water Usage Hypothesis :**

If real-time soil moisture data is available, then water usage for maintaining soil moisture can be optimized, reducing wastage and promoting sustainable agricultural practices.

▪ **Environmental Impact Hypothesis :**

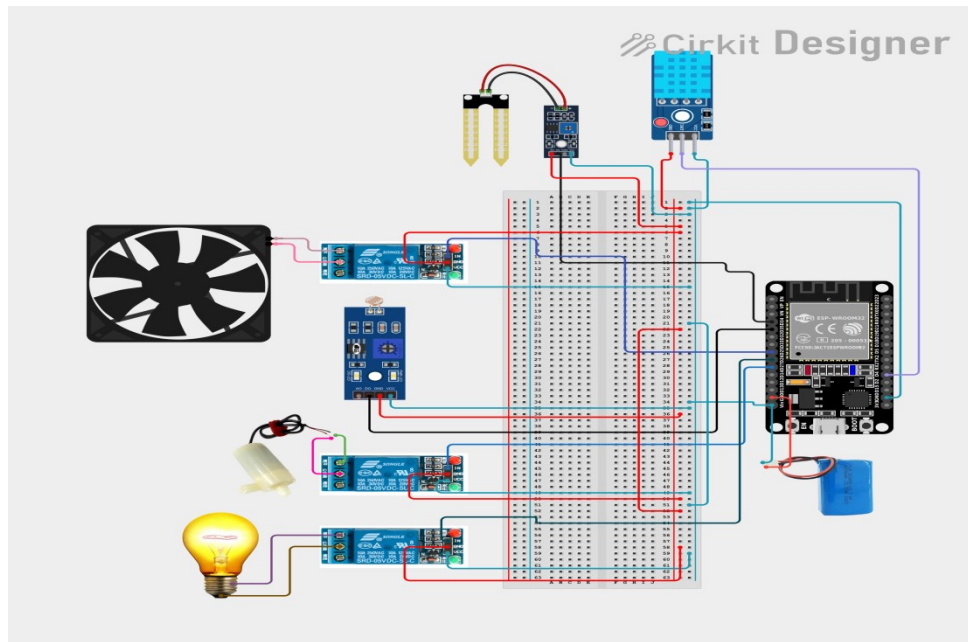
If greenhouse parameters are maintained within optimal ranges using real-time data, then plant stress due to environmental fluctuations can be minimized, leading to higher crop yield and lower resource consumption.

▪ **Scalability Hypothesis :**

If this prototype performs successfully, then it can be scaled to support larger greenhouses or integrated with automated control systems like irrigation pumps & fans.

METHODOLOGY

BLOCK DIAGRAM

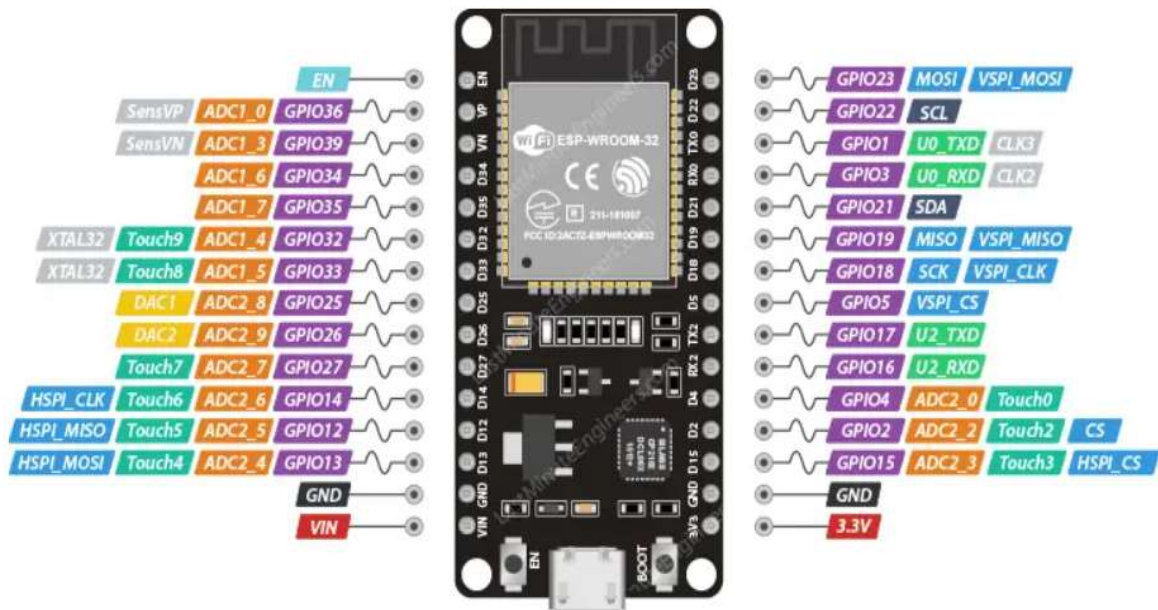


COMPONENTS REQUIRED

Component	Description	Quantity
ESP32 Microcontroller	Wi-Fi-enabled microcontroller for data processing and transmission.	1
DHT11 Sensor	Measures ambient temperature and humidity.	1
Soil Moisture Sensor	Detects moisture content in the soil.	1
LDR (Light Dependent Resistor) + 10k Ω Resistor	Measures light intensity inside the greenhouse.	1
Relay Module	Acts like switch to on / off the Fan, LED glow light and water sprayer	3
Breadboard, Jumper Wires	Used to connect all the components.	1
Power Supply (Battery/USB)	Supply power to the entire connections.	1
Fan, LED grow light, water sprayer (humidifier)	For monitoring the environmental parameters.	1 each

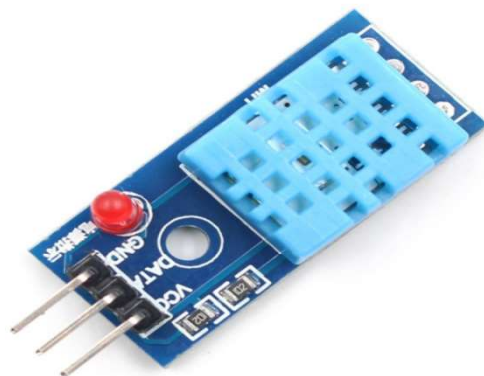
COMPONENTS BRIEF DESCRIPTION :

ESP32 :



The ESP32 is a popular, low-cost microcontroller series from [Espressif Systems](#) featuring integrated Wi-Fi, Bluetooth, and a dual-core processor, making it ideal for Internet of Things (IoT) applications. It includes a rich set of peripherals such as ADCs, DACs, and touch sensors, supports various programming languages (including C/C++, MicroPython, and JavaScript), and offers extensive community support. The ESP32 supports low-power modes and has a powerful clock speed of up to 240 MHz, providing a strong balance of performance, power efficiency, and versatility for a wide range of projects.

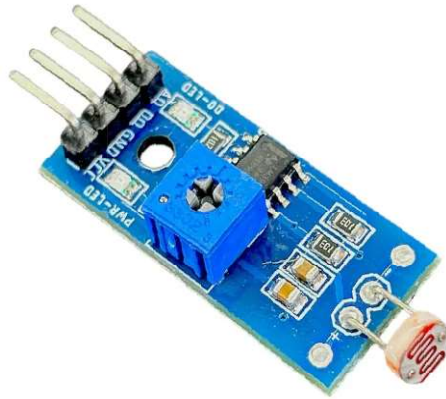
DHT11 SENSOR :



DHT11 is a low-cost digital sensor for sensing temperature and humidity. This sensor can be easily interfaced with any micro-controller such as Arduino, Raspberry Pi etc., to measure humidity and temperature instantaneously.

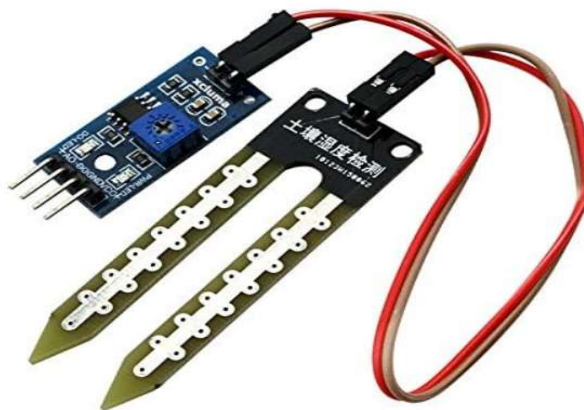
DHT11 humidity and temperature sensor is available as a sensor and as a module. The difference between this sensor and module is the pull-up resistor and a power-on LED. DHT11 is a relative humidity sensor. To measure the surrounding air this sensor uses a thermistor and a capacitive humidity sensor.

SOIL MOISTURE SENSOR :



A soil moisture sensor measures the water content in soil by detecting changes in electrical conductivity or other soil properties, with common types including resistive, capacitive, and Frequency_Domain_Reflectometry (FDR) sensors. These devices, consisting of probes and an interface module, are used for irrigation management in agriculture and home gardening, allowing for optimized water use, increased crop yields, and water conservation.

LDR :



An LDR sensor, or Light Dependent Resistor, is a passive electronic component that changes its electrical resistance based on the intensity of ambient light it's exposed to. The principle is photoconductivity, where the material becomes more conductive (resistance decreases) in brighter light and less conductive (resistance increases) in the dark. LDRs are simple, cost-effective, and widely used for applications like automatic streetlights, light-sensitive alarms, and in industrial automation to monitor and control light levels.

IMPLEMENTATION :

❖ HARDWARE IMPLEMENTATION :

1. Microcontroller:

- An ESP32 board serves as the brain of the system, with built-in Wi-Fi capabilities to connect to the internet.

2. Sensors:

- **DHT11 Sensor:** Measures both temperature and relative humidity in the greenhouse environment.
- **Soil Moisture Sensor:** Detects the moisture content in the soil, ensuring proper watering.
- **LDR (Light Dependent Resistor):** Measures the ambient light intensity to control the growing light.

3. Actuators:

- **Relay Module:** A relay module is used to control high-power devices like fans and lights using low-voltage signals from the ESP32.
- **DC Fan:** A DC fan is connected via a relay to cool the greenhouse when the temperature is too high.
- **Growing Light:** An LED grow light is connected via a relay to provide artificial light when the LDR detects low light intensity.
- **Water Sprayer/Pump:** A water pump or sprayer, connected to a relay, is activated to water the plants when soil moisture levels drop too low.

4. Connectivity & Power:

- **Wi-Fi Connectivity:** The ESP32's built-in Wi-Fi is used to send sensor data to the cloud or directly to the mobile app.
- **Power Supply:** A suitable power supply is needed for the ESP32, sensors, and actuators.

❖ SOFTWARE IMPLEMENTATION :

1. Programming Environment :

- **Include Libraries:** Include libraries for the DHT11 sensor, Wi-Fi, and communication with the mobile app platform (e.g., Blynk).
- **Sensor Reading:** Write code to initialize and read data from the DHT11 (temperature, humidity), soil moisture, and LDR sensors.
- **Wi-Fi Connection:** Configure the ESP32 to connect to your Wi-Fi network.
- **Actuator Control Logic:**

Temperature: If temperature > (e.g.) 28°C, turn on the fan; otherwise, turn it off.

Light: If light intensity < (e.g.) 100 lux, turn on the growing light; otherwise, turn it off.

Soil Moisture: If soil moisture < (e.g.) 40%, turn on the water sprayer; otherwise, turn it off.

- **Data Transmission:** Send the live sensor readings to the chosen mobile app platform (e.g., Blynk) using virtual pins.

2. IoT Dashboard :

- Blynk Mobile Application (iOS/Android) : Allows real-time monitoring of sensor values. Widgets like Value Display, Gauge, and Graph can be used. ESP32 connects to Blynk using Auth Token, Wi-Fi SSID, and Password.

3. Code Functionality :

- Initialization of all sensors and ESP32 Wi-Fi. Periodic reading of sensor values (temperature, humidity, soil moisture, light). Transmission of these values to the Blynk app via Wi-Fi. Optional : Alerts if values cross predefined thresholds (e.g., low moisture alert).

PROCEDURE :

1.Sensing the Environment :

- Multiple sensors are used to collect real-time environmental data:
- DHT11/DHT22 sensor – Measures temperature and humidity.
- Soil Moisture Sensor – Detects water content in the soil.
- LDR or Light Sensor – Measures light intensity.

2. Data Processing and Decision Making:

- The ESP32/ESP8266/Arduino UNO R3 reads all the sensor values and processes them. It continuously compares the readings with predefined threshold values.

Based on conditions:

- If temperature > threshold → Turn ON the fan to cool the greenhouse.
- If soil moisture < threshold → Turn ON the water sprayer to maintaining the soil moisture.
- If light intensity < threshold → Turn ON the LED grow light to supplement light.

3. Automation (Actuator Control):

The ESP32/ESP8266 controls relays connected to:

- Fan – for ventilation
- Water Sprayer – for maintaining soil moisture
- LED Grow Light – for light supplementation
- Each relay acts as a switch to control these external devices.

4. Wireless Communication (IoT):

- The ESP32/ESP8266 sends live sensor data via WiFi to a mobile app (Blynk). The user can monitor live environmental data and manually control devices from the app if needed.

CONNECTION DESCRIPTION :

- DHT11 :
VCC → 3.3V
GND → GND
Data → GPIO 4 (Digital)
- Soil Moisture Sensor :
VCC → 3.3V
GND → GND
Analog Output → GPIO 34 (Analog input)
- Fan (through Relay Module)
Relay VCC → 5V
Relay GND → GND
Relay IN → GPIO 25 (Digital Pin)
Fan → Connected to relay NO (Normally Open)
- LDR :
One end → 3.3V
Other end → 10k Ω resistor to GND and middle tap to GPIO 35 (Analog Input)
- ESP32 : It is powered via USB and connected to all the sensors and all other components through wires.
- Sprayer (piezo via humidifier) :
Connected to ESP32 of PIN number GPIO27 (Digital Output) through Relay Module.
- LED Grow Light :
Connected to ESP32 of PIN number GPIO26 (Digital Output) through Relay Module.

EXPECTED OUTCOME OF THE PROJECT :

1. Successful real-time monitoring of greenhouse environmental parameters:
 - Accurate sensing of temperature, humidity, soil moisture, and light intensity.
2. Automated control of key greenhouse devices based on sensor thresholds:
 - Fan turns on/off based on temperature/humidity levels.
 - LED grow light operates based on ambient light intensity.
 - Water pump activates automatically when soil is dry.
3. Stable and reliable wireless data transmission using ESP32/ESP8266 to a connected mobile application.
4. Live data display on a mobile app interface with clear visuals for all monitored parameters.
5. Reduction in manual labor and resource wastage, leading to improved efficiency in greenhouse management.
6. Improved plant growth and health due to optimal environmental conditions maintained automatically.
7. Enhanced user control and convenience with remote monitoring and potential manual override via mobile app. And Low-cost and scalable prototype, demonstrating a practical IoT-based solution for smart agriculture.

APPLICATIONS :

- ◆ Real-Time Environmental Monitoring and Control.
- ◆ Automated Smart Farming.
- ◆ Dual-Mode Operation (Manual and Automatic Control).
- ◆ Enhanced Resource Efficiency and Crop Yield.

ADVANTAGES :

- ◆ Real-Time Monitoring and Alerts
- ◆ Automated Environmental Control
- ◆ Dual Control Modes
- ◆ Energy and Resource Efficiency

DISADVANTAGES :

- ◆ Internet Dependency
- ◆ Limited Sensor Accuracy
- ◆ Initial Setup Complexity
- ◆ Maintenance and Calibration Needs

CONCLUSION :

The Smart Greenhouse Monitoring System successfully integrates IoT and mobile technology to provide a real-time solution for agricultural monitoring. By using ESP32 with low-cost sensors, this system addresses the shortcomings of previous systems that lacked remote access or were too expensive. Our solution is practical for small-scale farmers and hobbyists, offering both convenience and increased crop productivity. This project forms the basis for future expansions like automated irrigation or AI-based analytics.

REFERENCES :

D. Singh et al., “IoT-Based Smart Agriculture Monitoring System,” *International Journal of Scientific Research in Computer Science*, 2021.

G. S. Prakash, “Greenhouse Monitoring using Arduino and GSM,” *International Journal of Engineering and Advanced Technology*, 2020.

Blynk IoT Platform – <https://blynk.io>

ESP32 Documentation – <https://docs.espressif.com/projects/esp-idf/en/latest/esp32/>

Arduino IDE – <https://www.arduino.cc/en/software>

```
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT11

#define FAN_PIN 25
#define LIGHT_PIN 26
#define SPRAYER_PIN 27

#define SOIL_PIN 34
#define LDR_PIN 35

#define TEMP_THRESHOLD 30
#define MOISTURE_THRESHOLD 30
#define LIGHT_THRESHOLD 500

char auth[] = "Your_Blynk_Auth-Token";
char ssid[] = "Your_WiFi_SSID";
char pass[] = "Your_WiFi_PASSWORD";

DHT dht(DHTPIN, DHTTYPE);

bool autoMode = true;

void setup() {
  Serial.begin(115200);
  Blynk.begin(auth, ssid, pass);
```

```
dht.begin();

pinMode(FAN_PIN, OUTPUT);
pinMode(LIGHT_PIN, OUTPUT);
pinMode(SPAYER_PIN, OUTPUT);

digitalWrite(FAN_PIN, LOW);
digitalWrite(LIGHT_PIN, LOW);
digitalWrite(SPAYER_PIN, LOW);
}

// Auto/Manual mode toggle
BLYNK_WRITE(V4) {
    autoMode = param.asInt();
}

// Manual Control
BLYNK_WRITE(V5) { if (!autoMode) digitalWrite(FAN_PIN, param.asInt()); }
BLYNK_WRITE(V6) { if (!autoMode) digitalWrite(LIGHT_PIN, param.asInt()); }
BLYNK_WRITE(V7) { if (!autoMode) digitalWrite(SPAYER_PIN, param.asInt()); }

void loop() {
    Blynk.run();

    float temp = dht.readTemperature();
    float hum = dht.readHumidity();
    int soil = analogRead(SOIL_PIN);
    int light = analogRead(LDR_PIN);
```

```
// Map soil moisture (depends on your sensor)
int soilPercent = map(soil, 4095, 1500, 0, 100); // Adjust range based on calibration

Blynk.virtualWrite(V0, temp);
Blynk.virtualWrite(V1, hum);
Blynk.virtualWrite(V2, soilPercent);
Blynk.virtualWrite(V3, light);

if (autoMode) {
  digitalWrite(FAN_PIN, temp > TEMP_THRESHOLD ? HIGH : LOW);
  digitalWrite(LIGHT_PIN, light < LIGHT_THRESHOLD ? HIGH : LOW);
  digitalWrite(SPRAYER_PIN, soilPercent < MOISTURE_THRESHOLD ? HIGH : LOW);

  // Update widget states
  Blynk.virtualWrite(V5, temp > TEMP_THRESHOLD);
  Blynk.virtualWrite(V6, light < LIGHT_THRESHOLD);
  Blynk.virtualWrite(V7, soilPercent < MOISTURE_THRESHOLD);
}

delay(2000); // Read every 2 seconds
}
```